NEER BHARDWAJ

SENIOR SYSTEMS & AI ENGINEER

Master Technical Skills & Applied Engineering Systems Compendium

An exhaustive technical compendium indexing **49 core engineering skills, mathematical algorithms, and architectural patterns** engineered across 35+ custom production codebases, research implementations, and verified platforms. Demonstrates verified production mastery in **Autonomous AI Architectures & LangGraph Multi-Agent Runtimes**, **LLVM Compilers & Custom Language Toolchains**, **C++20 AR/VR Spatial Game Engines**, **Cryptographic Provenance Hash Chains & Zero-Knowledge Vaults**, **Computer Vision Biometrics**, and **Enterprise Full-Stack Platforms & Distributed Automation**.

- ⚡ Low-Level Systems & LLVM
- 🧠 Autonomous Cognitive AI & LangGraph
- 🕹️ C++20 Spatial Engine & ECS
- 🛡️ Cryptographic Provenance & Zero-Knowledge
- 👁️ 3D Computer Vision Biometrics
- 🌐 High-Throughput Web & Multi-Tenancy
- 🤖 Resilient Browser Automation & Scrapers

CORE ENGINEERING DISCIPLINES & COMPETENCIES MATRIX (49 TOTAL COMPETENCIES)

ENGINEERING DISCIPLINE	APPLIED COMPETENCIES & TECHNICAL STACK DEMONSTRATED IN SYSTEMS
1. Autonomous AI & Cognitive Systems	Cognitive Routing, Intent Decomposition, Autonomous Executive Controllers, Synthetic DNA Evolution, Multi-Tiered Memory, Bounded Safety Kernels, Quantized Inference, LangGraph Multi-Agent HITL Governance, Multilingual Regulatory Intelligence & Diffing, Multi-Model Unified Gateway.
2. Compilers, Language Design & Systems	Lexical Tokenization, Recursive Descent AST Parsing, LLVM IR Codegen, Static Type Systems, Custom Standard Library (`omlib`), SIMD Cache-Aligned Structs.
3. Spatial Computing, AR/VR & Game Engines	C++20 Entity Component System (ECS), High-Concurrency UDP Networking, Client Prediction, HLSL Shaders, Deterministic 3D Physics, Cross-Engine Bridging.
4. Computer Vision, Biometrics & ML	MediaPipe 3D Face Mesh, Eye Gaze Vector Estimation (EAR), Head Pose Telemetry (SolvePnP), NLP TF-IDF Threat Classifiers, ROC-AUC Anomaly Tuning.
5. Cybersecurity, Cryptography & Defensive	AES-256-GCM Zero-Knowledge Vaults, PBKDF2 Key Derivation, Shannon Entropy Auditing, Port/CVE Auditing, Tamper-Evident SHA-256 Product Passports & Provenance Chains, Hardened Authentication, Defensive Memory Safety.
6. OSINT, Graph Analytics & Intelligence	Open-Source Intelligence Footprinting, Recursive Multi-Hop Entity Discovery, Sentence-Transformers Temporal Event Reconstruction, Social Network Analysis (SNA), Louvain Clustered Topologies.
7. Full-Stack Web, APIs & Creative UI/UX	Next.js 14+ (SSR/SSG), React 18, Strict TypeScript, Multi-Tenant Relational PostgreSQL Architecture & RBAC, Voice-First & Camera-Driven PWAs, High-Throughput FastAPI Services, WebGL & Canvas Shaders.
8. Automation, Network Infra & DSP	Manifest v3 Virtual-DOM Stream Harvesters, Bézier Anti-Bot Automation, Self-Healing Playwright Orchestration, High-Concurrency API Tree Crawling & Backoff, Reversible Filesystem Categorization Engines, Rotating Proxies, MIDI Protocols.

🧠 Autonomous AI Systems & Cognitive Architectures

10 Competencies

Cognitive Intent Routing & Task Decomposition

`Python` `Intent Classifiers` `Task Trees` `Adaptive Compute` `Latency Budgets`

Engineered autonomous routing engines that dissect natural language queries into hierarchical task graphs. Dynamically provisions compute budgets and selects optimal reasoning modes (Fast vs. Multi-Step Deep Reasoning) based on algorithmic difficulty scores, eliminating token waste and maximizing latency efficiency for complex problem-solving workflows.

Autonomous Executive Controllers & Session Orchestration

`State Machines` `Execution Sandboxes` `Tool Calling` `Step Verification` `Audit Trails`

Architected deterministic executive controllers that govern agent lifecycles, execution sandboxes, and multi-tool orchestration. Implemented runtime step verification, hard timeout budgets, and rollback mechanisms, ensuring autonomous agents execute multi-step tool calls with zero uncontrolled side-effects and generate cryptographically verified audit trails of every reasoning step.

Synthetic Heuristic Evolution & Experience Distillation

`Genetic Algorithms` `Experience Distillation` `Checkpointing` `SHA-256 Gene Hashes`

Designed self-learning cognitive engines that analyze runtime execution traces, isolate successful reasoning heuristics, and distill them into persistent synthetic heuristic genes. Implemented hash-verified DNA evolution pipelines and state checkpointing, enabling autonomous systems to adapt strategies across sessions and continuously reduce repetitive problem-solving overhead.

Multi-Tiered Memory Systems (Episodic & Semantic)

`Vector Embeddings` `Semantic Retrieval` `SQLite` `Episodic Replay` `Context Gating`

Developed hybrid memory architectures unifying episodic execution logs, semantic vector retrieval, and procedural stores. Engineered memory gating and context-injection pipelines that retrieve historical interaction patterns and inject relevant contextual priors into live reasoning loops without overwhelming context windows or causing associative hallucination.

Continuous Daemon Scheduling & Energy Governance

`Async Daemons` `Background Scheduling` `Resource Governors` `OS Process Hooks`

Implemented continuous background runtimes and hardware-aware energy governors to orchestrate long-horizon autonomous tasks. Engineered resource-aware scheduling daemons that throttle computational throughput, monitor system utilization, dynamically adapt priorities during idle windows, and safely trigger maintenance cycles, state persistence, and memory compaction.

Bounded Safety Kernels & Operational Policy Enforcement

`Sandboxing` `RBAC` `Read-Only Guards` `Network Isolation` `Multi-Level Trust`

Engineered deterministic safety kernel layers that validate incoming commands, enforce active operational constraints (read-only access, network isolation, destructive-write guards), and restrict tool execution boundaries. Built multi-level trust verification mechanisms to ensure autonomous code execution adheres strictly to user-defined safety policies before applying system modifications.

Model Runtime Optimization & Local Inference Pipelines

`Model Registry` `Quantized GGUF/ONNX` `Fallback Cascades` `Response Streaming`

Built modular model registry and runtime abstraction layers capable of dispatching tasks across cloud APIs and quantized local inference endpoints. Implemented automatic fallback cascades, token rate-limiting, and response streaming protocols, ensuring uninterrupted model availability and high-throughput inference under volatile network conditions.

Human-in-the-Loop Multi-Agent Governance & Policy Gateways

`LangGraph` `StateGraph` `HITL Gateways` `Policy Verification` `SHA-256 Audit Trail`

Architected mission-critical multi-agent governance runtimes using LangGraph StateGraph pipelines. Implemented asynchronous Human-in-the-Loop (HITL) approval gateways with checkpointed pause/resume states, anti-hallucination uncertainty scoring, and SHA-256 hash-chained immutable audit ledgers, ensuring strict human authorization before consequential autonomous actions in production workflows.

Cross-Lingual Regulatory Intelligence & Obligation Extraction

`Regulatory NLP` `12+ Indic Languages` `Obligation Extraction` `Temporal Task Graphs` `Clause Diffing`

Engineered end-to-end regulatory compliance intelligence systems extracting statutory obligations across 12+ Indian languages. Developed cross-lingual semantic vector search, temporal compliance task graph generators, and semantic AST version diffing to identify clause revisions, cross-jurisdictional liabilities, and filing deadlines with zero manual review friction.

Multi-Model LLM Gateway & Provider Abstraction Runtimes

`Express.js` `Model Registry` `Encrypted Key Vaults` `Token Streaming` `Multi-Provider Failover`

Built a unified multi-provider AI model gateway consolidating heterogeneous LLM provider endpoints behind a standardized REST interface. Implemented local encrypted API key vaults, dynamic rate-limiting governance, automatic failover cascades across providers, and Server-Sent Events (SSE) token streaming for resilient, high-availability model orchestration.

Compilers, Language Design & Low-Level Systems

5 Competencies

Lexical Tokenization, Grammar & AST Parser Engineering

`Custom Lexer` `Recursive Descent Parsing` `Abstract Syntax Trees` `Syntax Diagnostics`

Designed custom programming language frontends featuring deterministic lexical analyzers and recursive descent AST parsers. Developed formal grammar specifications capable of processing custom syntax structures, operators, control flow statements, and nested block scopes while providing line-accurate compiler diagnostics, syntax error recovery, and structured AST representations.

LLVM Intermediate Representation (IR) & Native Codegen

`LLVM C++ API` `Target Machine Passes` `Native Code Emitters` `JIT Compilation`

Constructed compiler backends utilizing LLVM infrastructure to translate custom high-level Abstract Syntax Trees into optimized LLVM Intermediate Representation (IR). Implemented instruction emission, register allocation passes, target machine configurations, and native object code generation pipelines, producing high-performance standalone machine binaries from scratch.

Static Type Systems & Semantic Verification

`Symbol Tables` `Type Inference` `Scope Resolution` `Semantic Analysis Passes`

Implemented static type-checking and semantic verification engines for custom compiled languages. Designed lexical symbol tables, variable scope resolvers, and type coercion validators that enforce strict compile-time type safety, eliminate undefined symbol access, and optimize memory layout prior to native binary emission.

Custom Standard Library & Native Memory Primitives

`Manual Memory Mgmt` `I/O Buffers` `Core Math Primitives` `C ABI Interoperability`

Authored custom standard libraries and core runtime primitives to support native language execution. Engineered low-level memory allocation wrappers, string manipulation buffers, terminal I/O routines, and mathematical primitives. Implemented standard C ABI linking interfaces, enabling custom-compiled programs to invoke operating system system-calls and native external shared libraries.

SIMD Vectorization & Cache-Aligned Data Structures

`Data-Oriented Design` `Cache Line Alignment` `Memory Padding` `C++20` `SoA Layout`

Applied low-level hardware optimization techniques to maximize CPU cache locality and instruction throughput. Structured memory buffers along 64-byte cache lines, utilized struct-of-arrays (SoA) layouts to facilitate SIMD auto-vectorization, and eliminated pointer chasing in critical execution paths, achieving significant throughput speedups in compute-intensive loops.

Spatial Computing, AR/VR & Game Engine Architecture

5 Competencies

Entity Component System (ECS) Architecture

`C++20` `Data-Oriented Design` `Cache Locality` `Archetype Pooling` `Parallel Systems`

Architected high-performance, multithreaded Entity Component System (ECS) game server architectures in modern C++20. Applied data-oriented design principles, contiguous memory archetype storage, and component pooling to eliminate object-oriented cache misses, enabling real-time parallel system updates for thousands of simultaneous entities across high-frequency simulation ticks.

Low-Latency Network Synchronization & State Replication

`UDP Sockets` `Delta Compression` `Dead-Reckoning` `Client-Side Prediction`

Engineered authoritative client-server spatial networking systems built on raw UDP socket protocols. Implemented delta-compression algorithms, client-side movement prediction, and dead-reckoning interpolation to guarantee synchronized spatial positions across distributed clients, minimizing bandwidth utilization and eliminating visual stutter across variable network connections in multiplayer virtual environments.

HLSL Shader Programming & Unity Rendering Pipelines

`Unity C#` `HLSL` `Universal Render Pipeline (URP)` `Custom Shaders` `Spatial FX`

Developed custom HLSL vertex and fragment shaders integrated within Unity's Universal Render Pipeline (URP). Created interactive spatial visual effects, holographic depth cues, spatial boundary grids, and real-time lighting materials optimized specifically for high-frame-rate rendering across virtual and augmented reality head-mounted displays.

Cross-Platform Spatial Engine Bridging

`Protocol Translators` `TCP/WebSockets` `Minecraft & Roblox Bridges` `Serialization`

Engineered bidirectional cross-platform bridge protocols linking standalone C++ spatial simulation engines with external game platforms such as Minecraft and Roblox. Implemented serialization protocols, entity mapping wrappers, and real-time WebSocket state broadcasters, allowing spatial avatars and world modifications to mirror continuously across disparate gaming engines.

Deterministic 3D Spatial Physics & Collision Simulation

`AABB Bounding Boxes` `Raycasting` `Spatial Partitioning` `Fixed-Step Physics`

Implemented deterministic 3D physics engines featuring axis-aligned bounding box (AABB) collisions, raycasting, and spatial hashing grids. Maintained fixed-timestep simulation loops to ensure identical physics resolution across heterogeneous client hardware, preventing desynchronization in competitive spatial multi-user simulations.

Computer Vision, Biometrics & Machine Learning

4 Competencies

Facial Mesh & Eye Gaze Vector Estimation

`OpenCV` `MediaPipe Face Mesh` `Iris Tracking` `3D Geometry` `Euclidean Vectors`

Implemented real-time facial landmark tracking and gaze estimation pipelines using OpenCV and MediaPipe. Extracted 468 3D facial coordinate landmarks to calculate Euclidean Eye Aspect Ratios (EAR) and iris center displacements, accurately estimating user eye gaze direction and blink frequency at 60 FPS without specialized hardware sensors.

Biometric Attention & Posture Telemetry

`SolvePnP 3D Head Pose` `Real-Time Telemetry` `Ergonomic Analytics` `Fatigue Detection`

Built automated biometric tracking engines calculating 3D head pose rotation vectors (pitch, yaw, roll) using OpenCV's SolvePnP algorithm. Formulated attention scoring heuristics cross-referencing gaze stability, head orientation, and eye closure durations to detect distraction and fatigue, generating automated alerts and rich ergonomic productivity analytics.

NLP Feature Extraction & Threat Classification

`Scikit-Learn` `TF-IDF Vectorization` `N-Grams` `Regex Heuristics` `Ensembles`

Developed natural language processing pipelines to detect deceptive phishing emails and malicious messaging. Extracted statistical lexical features, suspicious URL structural patterns, domain spoofing heuristics, and TF-IDF word/character n-grams, applying ensemble machine learning classifiers to accurately categorize sophisticated social engineering attacks and automated spam campaigns.

Statistical Anomaly Detection & Model Evaluation

`Confusion Matrices` `ROC-AUC Curves` `Precision-Recall Tuning` `Scikit-Learn`

Engineered rigorous machine learning validation workflows evaluating model performance across imbalanced security datasets. Tuned classification thresholds, optimized ROC-AUC curves, and applied cross-validation to minimize false-positive rates in production security screening, ensuring legitimate high-priority communications remain unhindered.

Cybersecurity, Cryptography & Defensive Engineering

7 Competencies

AES-256-GCM Cryptography & Zero-Knowledge Vaults

`AES-256-GCM` `PBKDF2 HMAC-SHA256` `Secure Memory Scrubbing` `SQLite Security`

Architected zero-knowledge encrypted credential vaults utilizing authenticated AES-256-GCM encryption and PBKDF2 HMAC-SHA256 key derivation with high iteration counts and cryptographic salts. Implemented secure in-memory key handling with automatic memory scrubbing, tamper-resistant SQLite database encryption, and automated clipboard sanitation to prevent memory dump exploitation.

Password Strength & Cryptographic Entropy Calculation

`Shannon Entropy` `Zxcvbn` `Spatial Pattern Analysis` `Brute-Force Estimation`

Engineered real-time password security analyzers calculating information-theoretic Shannon entropy and heuristic pattern vulnerability. Evaluated dictionary permutations, spatial keyboard walks, sequential repetitions, and estimated offline brute-force cracking durations, delivering dynamic visual feedback and tailored suggestions to enforce resilient, entropy-dense passphrases resistant to modern GPU cracking.

Automated Port Auditing & CVE Vulnerability Mapping

`Socket Programming` `Multithreading` `Banner Grabbing` `CVE Databases` `Auditing`

Developed multi-threaded network reconnaissance tools in Python for automated infrastructure auditing. Performed concurrent TCP/UDP port scans, service identification, and raw banner grabbing, automating the cross-referencing of discovered daemon versions against published CVE vulnerability databases to deliver prioritized risk scores and actionable technical remediation summaries.

Hardened Authentication & API Gateway Security

`JWT / OAuth2` `Bcrypt Salted Hashing` `Rate Limiting` `RBAC` `CSRF/XSS Defense`

Engineered enterprise-grade authentication gateways featuring Bcrypt password hashing, short-lived JWT access tokens with rotating refresh tokens, and Redis-backed IP rate-limiting. Enforced strict HTTP-only secure cookie policies, CSRF token validation, and granular role-based access control (RBAC), effectively eliminating brute-force attacks and session hijacking vulnerabilities.

Custom Cryptographic Stream Obfuscation & Bitwise Ciphers

`Bitwise Permutation` `Dynamic S-Boxes` `Stream Ciphers` `Binary Masking`

Designed experimental cryptographic obfuscation algorithms and lightweight stream cipher prototypes. Implemented multi-round bitwise rotations, dynamic S-box substitution matrices, and key-dependent byte shuffling to obscure sensitive binary payloads, evaluating cipher resistance against statistical frequency analysis, pattern detection, and automated decompilation during static reverse-engineering analysis.

Defensive Memory Safety & Anti-Tamper Invariants

`Buffer Boundary Checks` `Secure Zeroing` `Cryptographic Hashes` `Integrity Checks`

Implemented defensive systems programming techniques that prevent buffer overflows, use-after-free conditions, and memory corruption. Built tamper-evident integrity checkers that continuously verify executable binary hashes, preventing code injection, DLL hijacking, and unauthorized runtime memory patching.

Cryptographic Product Passports & Tamper-Evident Provenance Chains

`SHA-256 Hash Chains` `Digital Passports` `Fair-Trade Ledgers` `QR/Shortcode Provenance` `Geospatial Anomaly`

Architected decentralized provenance platforms issuing cryptographically tamper-evident digital product passports. Engineered SHA-256 cryptographic hash chains linking physical item origin, raw material purity, and ethical compensation records, combined with dual QR/alphanumeric resolution and scan-velocity geospatial anomaly detection to neutralize counterfeit duplication attacks.

OSINT, Graph Analytics & Web Intelligence

4 Competencies

Open-Source Intelligence (OSINT) & Digital Footprinting

`Automated Scraping` `DNS Enumeration` `Metadata Mining` `Dossier Aggregation`

Developed modular OSINT intelligence tools automating entity footprinting across public digital channels. Scripted concurrent scrapers and API aggregators for multi-platform username validation, email deliverability probing, DNS domain tracking, and file metadata extraction, synthesizing disparate unstructured OSINT data points into unified investigation dossiers for security researchers.

Recursive Semantic Crawling & Knowledge Graphs

`Graph Theory` `Recursive Crawlers` `NetworkX` `D3.js` `Vector Nodes` `Topologies`

Engineered automated research engines that recursively crawl semantic web links to construct multi-dimensional knowledge graphs. Extracted named entities, clustered contextual relationships, and mapped topical hierarchies using graph theory data structures, enabling researchers to visualize complex conceptual domains, uncover hidden nodal links, and explore interconnected data visually.

Social Network Analysis (SNA) & Graph Centrality Metrics

`Eigenvector Centrality` `Betweenness` `Degree Centrality` `Louvain Community Detection`

Built network graph analytics platforms that reconstruct social communication structures and digital communities. Calculated eigenvector, degree, and betweenness centrality metrics to identify influential network nodes and information gatekeepers, applying Louvain community detection algorithms to segment topological clusters and analyze information propagation pathways across large datasets.

Recursive OSINT Entity Graph Discovery & Temporal Event Reconstruction

`Entity Tracing` `Sentence-Transformers` `Neo4j Graphs` `Wayback CDX` `Confidence Scoring`

Architected OSINT intelligence platforms that recursively trace digital identities across multi-level depth hops (1–5). Built confidence scoring engines evaluating relational link strengths, integrated Wayback Machine CDX temporal timelines, and utilized local Sentence-Transformers embeddings to chronologically reconstruct real-world geopolitical events and misinformation narratives from heterogeneous feeds.

💻 Full-Stack Web Engineering, APIs & Creative UI/UX

6 Competencies

Modern React, Next.js & Strict TypeScript Architecture

`React 18+` `Next.js SSR/SSG` `TypeScript` `Tailwind CSS` `Atomic Design Patterns`

Architected responsive, production-ready web applications using Next.js, React, and TypeScript. Implemented server-side rendering (SSR), static site generation (SSG), and atomic UI component design with Tailwind CSS, ensuring strict end-to-end type safety, modular state management, and optimized asset delivery for high-performance responsive user experiences.

High-Throughput Asynchronous Python APIs

`FastAPI` `Pydantic` `Uvicorn` `Asyncio Event Loops` `Connection Pooling` `OpenAPI`

Engineered high-concurrency backend services and RESTful API gateways using FastAPI and Pydantic. Leveraged asynchronous I/O loops (`asyncio`), database connection pooling, and strict schema validation models to deliver sub-millisecond response latencies under heavy traffic while automatically generating interactive OpenAPI (Swagger) documentation.

Multi-Tenant Relational Schema Architecture & Role-Based Isolation

`PostgreSQL 16` `Prisma ORM` `Tenant Isolation` `RBAC Enforcement` `Sales Pipeline State Machine`

Designed production multi-tenant database architectures featuring strict organizational tenant boundary isolation, role-based access control (Admin, Manager, Agent), and indexed PostgreSQL schemas via Prisma ORM. Engineered deterministic state machines, automated stage transition webhooks, and tamper-resistant audit activity logging across enterprise tenant environments.

Creative Web Development & Canvas Shader Animations

`Three.js` `WebGL Canvas` `GSAP ScrollTrigger` `99+ Lighthouse Optimization`

Built award-caliber creative web interfaces integrating custom WebGL canvas shaders, Three.js 3D scenes, and GSAP ScrollTrigger animations. Engineered buttery-smooth micro-interactions, responsive fluid typography, and asset preloading strategies, consistently achieving 99+ Google Lighthouse performance, accessibility, and SEO audit scores.

Adaptive Video Streaming & Media Optimization

`HLS / DASH Protocols` `CDN Edge Caching` `Video Transcoding` `Lazy Loading`

Built modern streaming frontends integrating adaptive bitrate video players, episode indexing, and dynamic metadata scrapers. Implemented CDN edge caching, video thumbnail generation, and client-side view state persistence, ensuring instant playback startup and seamless buffering across bandwidth-constrained mobile networks.

Voice-First & Camera-Driven Progressive Web Application Architecture

`Progressive Web App` `Web Speech API` `MediaStreams API` `Offline Service Worker` `Tailwind CSS`

Engineered camera-first, voice-guided Progressive Web Applications designed for accessible edge computing. Integrated browser-native Web Speech API synthesis/recognition for speech-driven navigation, real-time MediaStreams photographic provenance capture, and offline-first Service Worker caching, delivering resilient sub-second interaction under erratic network conditions.

⚙️ Automation, Browser Extensions, Network Infra & DSP

8 Competencies

Manifest V3 Service Workers & Virtual-DOM Stream Harvesters

`Chrome Extension MV3` `Service Workers` `Virtual DOM` `Client-Side CSV/JSON` `Asynchronous Storage`

Developed production Chrome extensions on Manifest V3 featuring background service worker coordination and high-frequency content script injection. Engineered memory-bounded virtual DOM mutation observers that harvest infinite scrolling feeds, sanitize media URLs and telemetry, and stream client-side multi-format exports (CSV, JSON, Link manifests) without memory leaks.

Humanized Behavioral Emulation & Anti-Bot Steganographic Navigation

`Selenium WebDriver` `Bézier Trajectories` `Micro-Jitter` `Session Persistence` `Contextual Ingestion`

Created human-indistinguishable browser automation engines employing non-linear Bézier cursor curves, variable scroll velocity physics, and randomized micro-jitter intervals. Implemented automated session cookie persistence, multi-condition DOM wait strategies, and anti-fingerprinting heuristics to safely bypass bot detection barriers on complex web platforms.

Multi-Egress Rotating Proxy & TLS Fingerprinting

`SOCKS5/HTTP Proxies` `JA3/JA4 Spoofing` `Asynchronous Socket Pools` `IP Rotation`

Built resilient network routing infrastructure managing dynamic pools of residential and datacenter proxies. Implemented automated IP rotation upon HTTP 429 throttling, health checking, and TLS client fingerprint spoofing (JA3/JA4 emulation), ensuring reliable, uninterrupted data acquisition across distributed web scraping and security assessment tasks.

Deterministic Filesystem Categorization & Reversible Relocation Engines

`Python 3` `MIME Inspection` `Collision-Safe Renaming` `JSON State Journals` `Tkinter GUI`

Developed high-speed local filesystem organization engines capable of non-destructively categorizing thousands of disparate files via MIME analysis and magic bytes. Built atomic collision-safe renaming algorithms (file (1).ext), transactional move journals (_organize_log.json), and instant one-click rollback engines that restore directory structures to their exact original state without data loss.

Digital Signal Processing (DSP) & MIDI Protocols

`TypeScript` `Web Audio API` `Polyphonic Synthesizers` `Binary MIDI 1.0/2.0 Protocols`

Developed comprehensive digital audio tools parsing binary MIDI 1.0/2.0 protocol specifications. Implemented raw byte decoding, multi-track timing clocks, velocity curve transformations, and real-time audio synthesis pipelines using the Web Audio API, delivering interactive piano-roll visualizations and high-fidelity polyphonic sound generation.

Generative AI Document & Presentation Automation

`python-pptx` `Structured Prompt Chains` `LLM Integration` `Layout Engines`

Engineered automated presentation synthesis pipelines utilizing LLM prompt chains and `python-pptx`. Programmatically generates structured slide hierarchies, contextual summaries, professional typography, and corporate color palettes directly from unstructured research documents, automating end-to-end slide deck generation without human manual formatting.

High-Concurrency API Tree Crawling & Adaptive Backoff Engines

`ThreadPoolExecutor` `GitHub REST API` `Exponential Backoff` `Binary Detection` `Windows Console TUI`

Built high-concurrency API crawlers and recursive repository tree replicators in Python. Utilized ThreadPoolExecutor worker pools, automated HTTP 429/403 rate-limit backoff cascades, binary payload stream discrimination, and interactive keyboard-driven Windows console TUIs (msvcrt) to clone deep nested directory hierarchies with zero thread deadlocks or packet loss.

Self-Healing Web Automation & Multi-Context Browser Orchestration

`Playwright Async` `DOM Healing` `Selector Cascades` `Browser Contexts` `Rate Limiting`

Engineered resilient browser automation pipelines equipped with self-healing selector fallback cascades to overcome dynamic DOM mutations. Architected isolated multi-profile browser context pooling, real-world page validation checkpoints, and operational rate-limiting cooldown protocols that mimic human engagement rhythms and prevent anti-scraping blocks.